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Region-Dependent Artificial Pain Perception Based on Synaptic Transistors Mimicking Differential Nociceptor Sensitivity

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**Region-Dependent Artificial Pain Perception Based on
Synaptic Transistors Mimicking Differential Nociceptor
Sensitivity**

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Abstract

Nociceptors, which detect pain against harmful stimuli, exhibit region-dependent sensitivity, leading to variations in perceived pain intensity across different parts of the body. Herein, we propose that region-dependent nociceptive responses emulate by reflecting differences in the number of activated fibers through low-pain-mimicking synaptic transistor (LST) and high-pain-mimicking synaptic transistor (HST), each characterized by distinct synaptic weights. The variation in the synaptic weight is achieved by modulating the charge-trapping capability of high-k HfO_x , which is precisely controlled by the ultraviolet treatment time. Under 64 consecutive spike inputs, the LST and HST exhibited average post-synaptic current variations of 134.0% and 261.0%, respectively, across device-to-device measurements. The back of the hand and the calf, which exhibit different pressure pain thresholds, were taken as representative examples to demonstrate that the perceived pain intensity differs across body regions in response to identical stimulus magnitudes. These results validate their capability to emulate region-dependent variations in pain perception across body regions. Furthermore, these results demonstrate that key nociceptive characteristics such as threshold behavior, no adaptation, and sensitization exhibit different firing points depending on the magnitude of the applied stimulus, and that the onset of these characteristics also varies with stimulus frequency. Furthermore, to more realistically emulate the nociceptive characteristics of the human body, a region-dependent artificial pain perception system was developed by integrating a force-sensitive resistor sensor, an Arduino-based signal processor, and synaptic transistors. Through this region-dependent artificial pain perception system, realistic pain-like sensory feedback was achieved while emulating sensitization phenomena such as allodynia and hyperalgesia, demonstrating a hardware-level implementation that reproduces region-dependent pain transmission observed in the human body and offers significant potential for neuroprosthetics, robotics, and e-skin applications.

Keywords: region-dependent, artificial pain perception system, nociceptor, hafnium oxide, ultra-violet treatment, synaptic transistor

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1. Introduction

Among various human sensory receptors, nociceptors are specialized receptors that perceive pain, serving as a vital component of the body's protective response by converting harmful mechanical, thermal, or chemical stimuli into a receptor potential, which is recognized as pain when it exceeds a threshold [1-2]. Accordingly, pain sensitivity varies across different body regions, leading to differences in the degree of pain perceived under identical external stimuli [3-5]. Consequently, the magnitude of the pain signal generated differs depending on the body region.

The variation in perceived pain sensitivity across different body regions arises from differences in factors such as regional innervation density, skin thickness, and subcutaneous fat distribution, which modulate the spatial summation of nociceptive inputs [6-8]. As a result, the number of activated nociceptive fibers can vary even under identical external stimuli, leading to region-dependent differences in perceived pain sensitivity. Accordingly, Figure 1(a) presents a schematic illustration of pain signal processing to visualize region-dependent differences in pain sensitivity arising from variations in the pressure pain threshold, which quantitatively reflect differences in human pain perception due to the number of activated fibers. Specifically, the back of the hand (mean threshold value: $196.2 \text{ N}\cdot\text{cm}^{-2}$) and the calf (mean threshold value: $84.4 \text{ N}\cdot\text{cm}^{-2}$) are highlighted [9]. When the receptor potential exceeds a certain threshold, an action potential is generated and propagated along the A δ and C nerve fibers toward the dorsal root ganglion (DRG), a neuronal cluster within the peripheral nervous system [10]. From the DRG, nociceptive signals are transmitted to the spinal cord, a principal component of the central nervous system, where they undergo synaptic relay and decussation to the contralateral side before ascending along nociceptive pathways toward higher brain regions involved in pain perception [11]. Variations in the number of activated fibers modulate the degree of neural activation and the amount of neurotransmitter release, thereby altering the strength of the transmitted pain signal. This modulated signal propagates through the spinal cord to higher brain regions involved in pain perception, including the thalamus, cerebral cortex, and amygdala, ultimately determining the perceived intensity of pain [12].

Recent advances in self-powered sensing platforms and synaptic devices have collectively driven the development of bio-inspired systems capable of efficiently acquiring, transmitting, and processing sensory information for applications in human-machine interactions [13-16]. These technological advances have further inspired hardware-level emulation of biological sensory functions, among which nociception has emerged as a particularly compelling target owing to its critical role in protective pain perception. Emulating nociceptive characteristics employs synaptic devices that enable simultaneous data storage and processing, yielding improved energy efficiency to overcome the bottleneck of conventional von Neumann computing [17-21]. Nociceptive behaviors, such as threshold, relaxation, no-adaptation, and sensitization, have been replicated through synaptic plasticity, where analog and gradual conductance modulation reflect nociceptor dynamics [22-24]. Recent studies have addressed pain modulation in nociceptors induced by various stimuli, such as light and heat. However, these studies have not considered the fact that nociceptors distributed across different body regions perceive different magnitudes of pain in response to the same stimulus, which limits their ability to realistically emulate nociceptors in the human body. To more realistically emulate nociceptors across different body regions, it is essential to incorporate sensory feedback that reflects regional differences in pressure pain thresholds, which arise from variations in the number of activated fibers and lead to differences in perceived pain intensity [25-28]. Reproducing these characteristics requires a three-terminal structure that allows wide-range, gate-controlled conductance modulation for differentiated responses [29]. To ensure the stable and reliable operation of such structures, inorganic materials provide a promising alternative to spin-coated, exfoliated, or electrolyte-based systems using two-dimensional materials and polymers, owing to their superior structural robustness, reproducibility, and stability [30-32].

Herein, we propose a synaptic transistor capable of realistically emulating nociceptive functions by reflecting variations in pain sensitivity arising from differences in the pressure pain threshold across body regions, thereby enabling the implementation of a

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region-dependent artificial pain perception system. The synaptic transistor operates through a charge-trapping mechanism within a three-terminal architecture. It offers broad tunability through an indium–gallium–zinc–oxide (IGZO) channel, chosen for its low off-state current and high scalability, and an inorganic-sputtered high- k HfO_x that functions as the charge-trapping layer [33–37]. To induce distinct variations in pain sensitivity, pain sensitivity layers (PSLs) were engineered by controlling the duration of ultraviolet (UV) treatment. Depending on the pressure pain threshold, both the presence or absence of pain perception and the magnitude of perceived pain in response to identical stimuli differ across body regions. The low-pain-mimicking synaptic transistor (LST) and high-pain-mimicking synaptic transistor (HST) were designed to emulate the nociceptive characteristics of the back of the hand and the calf, which exhibit a higher pressure pain threshold ($196.2 \text{ N}\cdot\text{cm}^{-2}$) and a lower pressure pain threshold ($84.4 \text{ N}\cdot\text{cm}^{-2}$), respectively, as schematically illustrated in Figure 1(b). A higher-pressure pain threshold indicates that fewer nociceptive fibers are activated by the input stimulus, resulting in lower pain sensitivity. Consequently, the calf (HST), which possesses a lower pressure pain threshold than the back of the hand (LST), generates a stronger signal in response to identical stimuli, thereby being perceived as a higher level of pain. The variations in the PSLs were analyzed by correlating charge-trap state formation with electrical characterization using current density–electric field (J–E) and capacitance–voltage (C–V) measurements, while photoluminescence (PL), X-ray reflectivity (XRR), and X-ray photoelectron spectroscopy (XPS) provided additional insight into the corresponding defect energy levels and bonding configurations. The synaptic characteristics of the LST and HST, which exhibit different sensitivities to stimuli, were further investigated by analyzing conductance variations, including modulation of the post-synaptic current (PSC) in response to changes in spike intensity and duration, as well as spike frequency-dependent plasticity (SFDP), long-term potentiation (LTP), and long-term depression (LTD). Building on this functionality, a region-dependent artificial pain perception system capable of implementing distinct pain sensitivity at the hardware level is proposed. Through this approach, the proposed system can deliver realistic sensory feedback for neuroprosthetics, robotics, and e-skin applications by emulating region-dependent nociceptive behavior.

2. Experimental section

2.1. Fabrication procedure of the LST and HST

A heavily boron-doped p-type silicon (p^+ -Si) with a thermally grown silicon dioxide (SiO_2) is adopted as the gate and gate insulator, respectively. Next, HfO_x was deposited using a radio frequency (RF) magnetron sputtering system under conditions of 20 standard cubic centimeters per minute (sccm) of argon (Ar) gas, a chamber pressure of 1 mTorr, and an RF power of 100 W for 8.5 minutes. After deposition, the sample underwent Ar plasma treatment at 100 W and 30 kHz for 5 minutes, followed by thermal annealing on a hot plate at 300 °C for 1 hour. Subsequently, the HfO_x was exposed to an ultraviolet (UV) lamp (wavelengths: 184.9 nm and 253.7 nm, power: 20 W) for durations of 10 minutes and 30 minutes, forming a low-pain-sensitivity layer and a high-pain-sensitivity layer (HPSL), respectively. Afterward, indium–gallium–zinc–oxide (IGZO) composed of In_2O_3 , Ga_2O_3 , and ZnO at a molar ratio of 1 : 1 : 1 was sputtered using an RF magnetron sputtering system under conditions of 60 sccm, 5 m Torr, and 150 W for 5 minutes, with a shadow mask defining a channel width by length (W/L) of 1 000 μm /150 μm . Thermal annealing was performed on a hot plate at 150 °C for 1 hour following IGZO deposition. Finally, an aluminum (Al) electrode was fabricated via RF magnetron sputtering under sequential conditions of 20 sccm and 5 mTorr, initially at 60 W for 10 minutes and subsequently at 140 W for 20 minutes.

2.2. Fabrication procedure of HfO_x , LPSL, HPSL-based MIM and MOS capacitors

Metal–insulator–metal (MIM) and metal–oxide–semiconductor (MOS) structures were fabricated through an identical process, with the only difference being the use of a p^+ -type silicon substrate for the metal role in the MIM and a lightly boron-doped p-type silicon for the semiconductor role in the MOS. The deposition of HfO_x , Ar plasma treatment, and thermal annealing were carried out in the same manner as the fabrication procedure for the LST and HST devices. After the deposition of HfO_x , LST, and HST, the Al electrode was deposited through a shadow mask with a circular opening of 0.09 mm radius using RF magnetron sputtering under the same conditions as previously described.

2.3. Fabrication procedure of the Sputter and ALD- HfO_x film.

ALD- HfO_x with a thickness of approximately 30 nm was deposited using tetrakis(ethylmethyamido) (TEMAHf) as the hafnium precursor and O_3 as the oxidant. The deposition was carried out at a chamber temperature of 280 °C, and the growth per cycle (GPC) of HfO_2 was maintained at ~0.12 nm per cycle. Sputter- HfO_x with a thickness of approximately 30 nm was deposited under identical conditions for the fabrication of the LST and HST for 9 minutes and annealed on a hot plate for 1 h at 280 °C.

2.4. Film and device characterization

The electrical characteristics of the synaptic transistors were measured using a parameter analyzer (4200A, Keithley Instruments, USA) under dark conditions. Additionally, capacitance–voltage (C–V) measurements of MOS structures were performed using an LCR meter (4284A, Hewlett-Packard, USA) with a frequency range of 20 Hz to 1 MHz. The cross-sectional device structures of the synaptic transistors were investigated by high-resolution transmission electron microscopy (HR-TEM) (JEM-ARM 200F NEOARM, JEOL). To examine the UV-treatment-dependent surface morphology of the HfO_x films, atomic force microscopy (AFM) (NX-10, Park Systems) was employed. Changes in oxygen vacancy-related energy levels as a function of UV treatment time were evaluated by photoluminescence spectroscopy (PL) (FS5, Edinburgh Instruments). Film density variations in the HfO_x were analyzed using X-ray reflectivity (XRR) (SmartLab, Rigaku), and the chemical bonding states between Hf and O were characterized by X-ray photoelectron spectroscopy (XPS) (K-alpha, Thermo Scientific, UK). Chemical bond configurations were analyzed by Fourier transform infrared spectroscopy (FTIR) (INVENIO, Bruker) for sputter- and ALD-deposited HfO_x , as well as for sputter-deposited HfO_x subjected to varying UV treatment durations of 10, 20, and 30 min. To qualitatively compare the charge trap states

formed in the LPSL and HPSL, time-resolved photoluminescence (TRPL) analysis was performed (Fluorolog-QM, Horiba) by comparing the carrier lifetime of each sample. The absorption characteristics of the UV light source with respect to HfO_x were assessed using UV–visible spectrophotometry (UV–vis) (V-650, JASCO). Additionally, the variation in electric field distribution within the HfO_x according to UV treatment duration was simulated using technology computer-aided design (TCAD) (Victory Omni 2D, Silvaco).

2.5. Electrical signal conversion process of the region-dependent artificial pain perception system

A force-sensitive resistor (FSR) 406 sensor was employed to detect external mechanical stimuli corresponding to applied weights of 50, 100, and 150 g. FSR 406 has a measurable weight range of 10–1 020 g, with a repeatability of $\pm 2\%$ and a hysteresis of $\pm 5\%$, ensuring reliable and consistent detection of the applied stimuli. The FSR was connected to a microcontroller unit (Arduino UNO R3), which converted the sensor's resistance change into a corresponding voltage signal. This voltage signal was transmitted to a digital-to-analog converter (DAC, AD5683RBRMZ-3-RL7), which produced an analog output range of 0–5V. The output was subsequently amplified using a tower-mounted amplifier (TMA, Model: TMA 0515D) to ± 15 V. The amplified signal was then converted into a bipolar voltage of ± 15 V using an operational amplifier (OPA) with a gain of 6 and applied as a pulse to the gate terminal of the synaptic transistor to emulate external nociceptive stimuli. The source terminal of the device was grounded with the Arduino, while the drain terminal was connected to a parameter analyzer for measurement. The post-synaptic current (PSC) was measured as a function of applied weight, enabling the analysis of synaptic response characteristics in relation to stimulus intensity.

3. Results and discussion

3.1. Charge-trapping characteristics of UV-treated HfO_x -based synaptic transistors

The schematic fabrication process of the PSL-based synaptic transistor is illustrated in Figure 2(a). To modulate the charge trap density of HfO_x , the UV treatment was adjusted to 10 and 30 minutes, resulting in the formation of the low-pain-sensitivity layer (LPSL) and high-sensitivity layer (HPSL), respectively. The device structure of the PSL-based synaptic transistor is further confirmed by the cross-sectional high-resolution transmission electron microscopy (HR-TEM) image shown in Figure 2(b). The image distinctly shows well-defined and continuous interfaces between each layer, revealing that the PSL and IGZO channels responsible for charge trap behavior have thicknesses of approximately 36.3 and 24.6 nm, respectively.

To emulate region-dependent pain perception, PSL-based synaptic transistors are designed to generate different magnitudes of action potentials in response to identical stimuli, thereby reproducing varying pain intensities. This behavior, which perceives different pain intensities, is emulated by tuning the charge-trapping capability of the PSL through conductance modulation. However, PSL composed of HfO_x deposited by sputtering, where high-energy ion bombardment induces nonuniform film growth, increases surface roughness that promotes trap formation at the channel interface and consequently degrades overall device performance [38–41]. To overcome these issues, argon (Ar) plasma treatment was employed to enhance both the electrical properties and surface morphology of the sputter-deposited HfO_x , as demonstrated by the hysteresis characteristics shown in Figure S1 and Table S1, and the surface morphology observed in the atomic force microscopy (AFM) images is shown in Figure S2.

To investigate the variation in the charge-trapping capability of HfO_x with respect to UV treatment, UV light with wavelengths of 184.9 and 253.7 nm was applied under a fixed power of 20 W, with the treatment time precisely adjusted as the sole variable parameter. The degree of charge-trapping capability under different UV treatment times was evaluated by measuring the hysteresis characteristics under bidirectional gate voltage (V_G) sweeps of ± 20 V at a fixed drain voltage of 10.1 V. As shown in Figure S3, the devices exhibited clockwise hysteresis loops, indicating the influence of charge-trapping effects [42–45]. The variations in the threshold voltage shift (ΔV_{TH}), subthreshold swing (S.S.), and $I_{ON/OFF}$, which are factors associated with charge-trapping behavior,

extracted from the hysteresis curves as a function of the UV treatment time, are presented in Figure 2(c), with the corresponding numerical data summarized in Table S2.

As the UV treatment time increased, both the ΔV_{TH} and S.S. values increased, whereas the $I_{ON/OFF}$ decreased, indicating an increase in the charge-trapping capability of HfO_x . These variations are attributed to electron trapping in the HfO_x originating from the IGZO channel. When the UV treatment time exceeded 30 minutes, the increase in ΔV_{TH} , representing the charge-trapping capacity, became less significant and approached saturation, indicating a limit to additional charge trapping. Conversely, a sharp increase in the S.S. value and a noticeable decrease in the $I_{ON/OFF}$ were observed. The increase in the S.S. value indicates an increase in the interface trap density (D_{IT}) at the HfO_x /IGZO interface, which is likely attributed to the deterioration of the surface morphology of HfO_x [46]. To further support this observation, AFM analysis was conducted to evaluate the changes in the surface roughness of HfO_x induced by UV treatment. As presented in Figure S4, the root-mean-square roughness (R_{rms}) values of 0.249 and 0.291 nm for the 30- and 40-minute UV-treated HfO_x , respectively, indicate that the surface roughness increases with longer UV treatment duration. Furthermore, the sharp decline in the $I_{ON/OFF}$ is attributed to trap-induced screening caused by excessive electron trapping in the HfO_x , which reduces the gate electric field and consequently limits electron accumulation in the IGZO channel, leading to degraded switching performance of the transistor [47-49]. Overall, when the UV treatment duration exceeds 30 minutes, charge trap-related states are formed to an extent that significantly contributes to the deterioration of transistor performance, as evidenced by the substantial degradation in the S.S. and $I_{ON/OFF}$ characteristics, which indicates that the device is no longer suitable for reliable transistor operation.

Based on these results, achieving region-dependent pain perception requires distinctive characteristics that reflect differences in charge trap capability. To differentiate variations in pain perception, HfO_x subjected to 10- and 30-minute UV treatments were selected because they exhibited a sufficiently large ΔV_{TH} difference and stable transistor operation, which is required for region-dependent pain perception. In contrast, the 5-minute UV-treated condition was excluded because its charge-trapping capability showed only a marginal improvement over the UV-untreated condition, making it inadequate for achieving the desired level of charge trapping. Schematic illustrations of the devices and their corresponding hysteresis characteristics are shown in Figure 2(d) and (e), where the 10-minute UV-treated HfO_x was designated as LST with a ΔV_{TH} of 3.33 V, and the 30-minute UV-treated HfO_x was designated as HST, with a ΔV_{TH} of 9.33 V. Furthermore, to validate the HfO_x thickness employed in this study, the optimized condition, the hysteresis characteristics and corresponding ΔV_{TH} values of IGZO transistors with varying HfO_x thicknesses under UV treatment durations of 10 and 30 minutes are provided in Figure S5. These results confirm that the HfO_x thickness adopted for both the LST and HST constitutes the optimized condition for achieving reliable charge-trapping efficiency while maintaining stable transistor operation.

To further assess the stability and reproducibility of the LST and HST, device-to-device variability and cycle-to-cycle endurance characteristics were evaluated across ten devices and fifteen cycles, respectively, as presented in Figure S6. The statistical results of the mean value and standard deviation confirm that both devices exhibit stable and repeatable performance, as shown in Figure 2(f). Furthermore, to evaluate the long-term stability of the LST and HST, the hysteresis characteristics were measured over 5 consecutive cycles at 1 day and 30 days after fabrication. The corresponding data are presented in Figure S7, where the marginal variation observed over time in ΔV_{TH} demonstrates that both devices maintain a stable charge-trapping capability over extended periods, confirming their long-term reliability. The equipment employed in our device fabrication process includes RF sputtering, plasma treatment, and UV treatment. Since these tools are already in use in the display and semiconductor industries, there is sufficient potential for scalability by customizing them to meet the requirements of this study [50-56].

To clarify the origin of the ΔV_{TH} variation induced by different UV treatment times, technology computer-aided design (TCAD) simulations were conducted. The simulated hysteresis characteristics closely matched the experimental results, and the

corresponding electric field distributions at a V_G of 20 V are shown in Figure S8. The simulations reveal that the electric field within the UV-untreated HfO_x layer (UUHL), LPSL, and HPSL progressively decreases with longer UV treatment durations due to increased electron trapping in the HfO_x under positive gate bias, which screens the gate field. The color mapping transition from red to green in the simulation further supports that the larger ΔV_{TH} observed in the HPSL originates from its stronger charge-trapping capability. These configurations demonstrate that controlled UV treatment duration enables tuning of the charge trap density, providing a reliable means to implement synaptic transistors with distinct pain sensitivity characteristics to mimic the region-dependent pain perception system.

3.2. Distinct synaptic characteristics of the LST and HST under different conductance variations

To emulate distinct pain sensitivities arising from differences in the number of activated nociceptive fibers across body regions, the output response should vary under identical input stimuli. For this purpose, the LST and HST were fabricated with a three-terminal architecture, in which the gate functions as the presynaptic terminal and the source/drain serves as the postsynaptic terminal to mimic synaptic behavior [57]. The difference in charge-trapping capability between the LST and HST enables modulation of the IGZO channel conductance, thereby reproducing different synaptic behaviors that result in different pain sensitivities.

Synaptic behavior based on charge trapping was examined by observing the post-synaptic current (PSC) variations of the IGZO channel in the LST and HST under positive gate spikes. The variations in the PSCs of the LST and HST were examined as a function of spike intensity, as shown in Figure 3(a)–(b) respectively, where gate spikes of 10, 11, 12, 13, and 14 V were applied with a drain voltage of 0.5 V and a spike duration of 10 ms. With increasing spike intensity, a stronger gate electric field is applied to the IGZO channel, which increases channel conductance and consequently results in higher PSC values for both devices. The LST exhibits a higher PSC than the HST due to its lower degree of electron trapping, allowing a greater number of accumulated electrons in the channel. To examine the residual current after the 10 ms spike pulse, the 20–40 ms range was magnified in the right-side graph, revealing that the PSC decreases with increasing spike intensity. The HST shows a lower residual current compared to the LST, indicating a larger amount of trapped charge, which induces a stronger gate-field screening effect and reduces channel conductance. These results demonstrate that modulation of trap density enables the emulation of distinct synaptic responses under identical spike intensities. This behavior resembles that of synaptic hyperpolarization in biological systems, where negatively charged ions such as Cl^- lower the membrane potential below the resting level (approximately -70 mV), thereby suppressing synaptic plasticity and regulating signal transmission [58]. Additionally, Figure S9 presents the PSC variation in the LST and HST as a function of spike duration within a fixed spike intensity of 14 V, exhibiting a trend consistent with spike intensity-dependent behavior. As the spike duration increases, a greater amount of charge trapping occurs, resulting in more significant PSC modulation. To evaluate the reliability of the LST and HST characteristics, we assessed the device-to-device variability by measuring 10 devices under a gate spike condition of -10 V. As shown in Figure S10, the devices exhibit robust characteristics with negligible variation, confirming the consistent and reproducible performance across multiple devices.

Furthermore, to evaluate whether the modulation of synaptic connection strength enables consistent variation in synaptic weight, the characteristics of SFDP were investigated. Synaptic weight refers to the modulation of connection strength between neurons induced by presynaptic stimulation in biological neural systems. From a device perspective, the synaptic weight is defined as the variation in the PSC in response to applied input stimuli, reflecting the degree of conductance modulation of the synaptic device [59]. The LST and HST were tested by applying pulse trains at frequencies of 20, 30, 40, and 50 Hz under a read voltage of 2.5 V, with a spike intensity of 14 V and a duration of 5 ms, as shown in Figure 3(c). This analysis examines how effectively the synaptic weight adapts when pain-related stimuli are repeatedly applied. As the frequency increased to 20–50 Hz, both the LST and HST exhibited a gradual reduction in PSC, showing that the difference in the synaptic response increased at higher frequencies. This behavior can be attributed to the shorter interval between consecutive spike pulses, which leads to a greater amount of trapped

charge in the devices. The PSC variations before and after pulse application at different frequencies are compared in Figure 3(d), which clearly shows that the change in the PSC for the HST is greater than that for the LST. This implies that the HST accumulates more trapped charge, resulting in a greater change in the PSC. Consequently, the gap in the synaptic response between the two devices becomes larger, reflecting their distinct charge-trapping characteristics and synaptic weight behavior.

While SFDP reflects short-term synaptic adaptation to repeated stimuli, LTP and LTD were further analyzed to evaluate the persistent modulation of synaptic weight, representing the strengthening and weakening of synaptic connections under repeated stimulation. These characteristics were measured, as shown in Figures 3(e) and (f), by applying 64 consecutive negative spikes (−14 V, 30 ms) for potentiation and 64 positive spikes (+12 V, 10 ms) for depression under a read voltage of 2.5 V. Potentiation and depression were implemented through the modulation of channel conductance, where potentiation corresponds to an increase in conductance caused by electron de-trapping from the PSL, and depression corresponds to a decrease in conductance due to electron trapping. However, a higher intensity and longer duration of negative spikes were required for potentiation, as the n-type IGZO channel lacks holes as minority carriers, resulting in inefficient hole injection and making electron de-trapping difficult to achieve under the same spike pulse conditions as positive spikes [60]. The LST exhibited a normalized conductance change from 64.5 to 87.1 nS, whereas the HST showed a larger modulation from 19.7 to 52.0 nS, indicating greater conductance tunability. This enhanced modulation in the HST is attributed to its higher density of electron trap sites, which facilitates more significant conductance variation under identical stimuli. As shown in Figure 3(e) and (f), 64 consecutive potentiation and depression pulses resulted in PSC increases of 134.1% for the LST and 264.1% for the HST.

$$\text{Dynamic range} = \frac{G_{\text{MAX}}}{G_{\text{MIN}}} \tag{1}$$

$$E = \text{PSC} \times V_{\text{DS}} \times t_{\text{DURATION}} \tag{2}$$

Accordingly, the dynamic range, defined as the ratio of the maximum to minimum conductance, is determined by Equation 1, which was calculated to be 1.35 for the LST and 2.64 for the HST [61]. The energy consumption per spike was estimated by Equation 2, where PSC denotes the post-synaptic current, V_{DS} is the drain voltage, and t_{DURATION} represents the spike width [62]. The PSC values were measured to be 985.4 nA for the LST and 466.0 nA for the HST, with spike durations of 10 ms for potentiation and 30 ms for depression at a fixed drain voltage of 0.5 V. Based on these parameters, the energy consumption per spike of the LST was calculated as 14.78×10^{-9} J for potentiation and 4.92×10^{-9} J for depression, whereas the HST exhibited lower values of 6.99×10^{-9} J and 2.33×10^{-9} J, respectively. These results confirm that differences in trap density led to distinct synaptic weight modulation behaviors, demonstrating the feasibility of emulating region-dependent pain perception. In addition, the reproducibility of the synaptic characteristics of the LST and HST was systematically evaluated through both device-to-device variability and cycle-to-cycle endurance analyses. The LTP/LTD behaviors were examined across ten different devices to assess device-to-device variation, followed by cycle-to-cycle endurance measurements performed on the same device to verify the stability of synaptic operation under repeated stimulation. A total of 1 024 consecutive pulses were applied during these evaluations, and the corresponding results are summarized in Figure S11. Based on the statistical results in Figure 3(g), the mean value and standard deviation of the dynamic range confirm that both devices exhibit stable and repeatable performance when they are adopted as a synaptic transistor. These analyses demonstrate that both the LST and HST exhibit stable and reproducible synaptic characteristics across multiple devices and repeated operating cycles. However, the LST and HST devices require relatively high gate voltages, as the ~120 nm thermalized grown SiO₂ gate insulator limits the electric field across the charge-trapping layers (LPSL and HPSL) and the IGZO channel at lower voltages. Reducing the SiO₂ thickness would enable an equivalent electric field to be achieved at lower applied voltages, facilitating low-voltage device operation [63]. Furthermore, the adoption of a flexible substrate such as polyimide is expected to significantly enhance the mechanical flexibility of the device, broadening its potential for application in wearable and implantable neuroprosthetic systems [64].

3.3. Analysis of charge-trapping capability variations in HfO_x with different UV treatment times

The modulation of the synaptic weight with varying UV treatment time in HfO_x arises from differences in the charge storage capacity between the LPSL and HPSL, which are determined by the alteration of the charge trap states. To elucidate the variation in charge trap states, the insulating and dielectric properties of HfO_x subjected to different UV treatment durations were analyzed to confirm the presence of trap-related states and to determine the energy levels at which these states are formed. Furthermore, the chemical bonding configurations within HfO_x were examined to identify the bonding changes responsible for the formation of these trap states.

The insulating characteristics were evaluated by J–E curves obtained from metal–insulator–metal (MIM) capacitors based on a UUHL, LPSL, and HPSL, as shown in Figure 4(a). The breakdown electric field (E_{BR}) of UUHL, LPSL, and HPSL was determined to be 5.25, 4.68, and 3.89 $\text{MV} \cdot \text{cm}^{-1}$, respectively, exhibiting a decrease with increasing UV treatment duration. This observed reduction in E_{BR} is attributed to trap-assisted tunneling, which promotes the formation of leakage current paths, while prolonged UV treatment leads to an increased density of interface trap states in HfO_x [65–66]. To further investigate the effect of UV treatment on the dielectric behavior, C–V measurements were conducted using MIM capacitors at frequencies of 1, 10, 100 kHz, and 1 MHz, and metal–oxide–semiconductor (MOS) capacitors at 100 kHz, as presented in Figures 4(b) and S12. With increasing UV treatment duration, a decrease in the dielectric constant was observed, attributed to a decrease in polarization of the HfO_x under an applied electric field, which resulted in a decrease in the capacitance. The dielectric constants of UUHL, LPSL, and HPSL were calculated at 1 MHz, yielding values of 11.30, 10.59, and 9.07, respectively.

$$D_{it} = \left(\frac{S.S.\log(e)}{kT} - 1 \right) \frac{C_{ox}}{q} \quad (3)$$

Based on the extracted capacitance values, D_{IT} was calculated by Equation 3 using the S.S. for the UV-untreated HfO_x synaptic transistor (UST), LST, and HST. The corresponding D_{IT} values were determined to be 6.99×10^{12} , 7.64×10^{12} , and $7.86 \times 10^{12} \text{ cm}^{-2}$, respectively, indicating that the number of interface traps progressively increases with prolonged UV exposure [56, 67–68]. Additionally, normalized C–V curves were analyzed to identify the underlying factors contributing to these dielectric variations. With prolonged UV treatment, the flat-band voltage (V_{FB}) exhibited a negative shift accompanied by a gradual decrease in capacitance, indicating the increased formation of positively charged species. This behavior suggests that charge trapping occurs not only at the interface but also within the bulk region of HfO_x [69–70]. These results confirm that the formation of positively charged species capable of acting as trap states increases with prolonged UV treatment.

To examine the variation in bulk trap states and determine the energy levels at which these traps are located, PL analysis was performed on UUHL, LPSL, and HPSL subjected to different UV treatment conditions. As presented in Figure S13, the photoluminescence excitation (PLE) spectra of the samples reveal comparable excitation energies of approximately 5.20 eV, along with similar PL intensities [71–72]. These observations indicate that electron excitation occurs at energy levels and intensities in the samples, which corresponds to the bandgap of the HfO_x -based dielectrics. However, the PL emission spectra presented in Figure 4(c) exhibit distinct differences in PL intensity at a photon energy of 2.52 eV, corresponding to energy levels commonly associated with defect states arising from oxygen vacancy (V_o) within the bandgap of HfO_x [73]. To provide a more quantitative comparison of the PL peak intensities among UUHL, LPSL, and HPSL, the PL spectra were integrated over the 480–505 nm range, which encompasses the sharp characteristic peak at 492 nm, as shown in Figure S14. The integrated intensities reveal that compared with UUHL, LPSL and HPSL exhibit 2.29- and 3.38-fold higher charge trap-related states, respectively, indicating a progressive increase in the trap density. To more quantitatively assess the charge trap-related states in the LPSL and HPSL samples, time-resolved photoluminescence (TRPL) analysis was performed, with the corresponding data and a detailed description provided in Figure S15. The extracted carrier lifetimes were 2.29 ns for LPSL and 5.08 ns for HPSL, with the notably longer lifetime observed in HPSL, confirming the presence of a higher density of charge trap-related states compared to LPSL [74–75]. An increase in the PL intensity

at 2.52 eV, which corresponds to V_O -related emission, indicates a higher concentration of oxygen vacancies, dangling bonds, and multiple oxygen-related species, suggesting that prolonged UV treatment leads to increased formation of charge trap states within HfO_x [76-79]. The increase in the concentration of charge trap-related states with longer UV treatment indicates a change in the Hf-O bonding state within the HfO_x , which can in turn affect the overall film density. Accordingly, XRR analysis was performed to examine the UV-induced changes in thin-film density, revealing that the film densities of UUHL, LPSL, and HPSL were 8.04, 7.99, and $7.93 \text{ g} \cdot \text{cm}^{-3}$, respectively, as shown in Figure S16. As the film density is proportional to the critical angle, a slight decrease in the critical angle with increasing UV treatment duration was observed, indicating a marginal reduction in film density under prolonged UV exposure. This variation in film density is not attributed to factors such as crystallization, as the annealing temperature of 300°C is insufficient to promote crystallization in HfO_x , but rather reflects changes in the chemical bonding configuration between metal and oxygen atoms [80-81].

To examine the variation in chemical bonding states under different UV treatment durations, XPS analysis was conducted. The deconvoluted Hf 4f XPS spectra of the bulk regions for UUHL, LPSL, and HPSL are presented in Figure 4(d)-(f), respectively, while the corresponding interface regions can be found in Figure S17. Hf 4f peaks exhibit spin-orbit splitting, with Hf 4f^{7/2} and Hf 4f^{5/2} associated with Hf-O bond of binding energy (B.E.) at $(17.6 \pm 0.2) \text{ eV}$ and $(19.2 \pm 0.2) \text{ eV}$, respectively, while the peak at B.E. $(15.9 \pm 0.2) \text{ eV}$ is attributed to sub-oxides [82-83]. These suboxide peaks are associated with uncoordinated oxygen species such as V_O , and hydroxyl groups are formed by non-stoichiometric Hf-O bonds that generate charge trap states within the film [84-85]. XPS analysis revealed that the suboxide spectral area in the bulk region of UUHL was approximately 4.1%, while those of LPSL and HPSL increased to 10.8% and 14.8%, respectively. Similarly, in the interface region, the suboxide spectral areas of UUHL, LPSL, and HPSL were determined to be 5.6%, 11.1%, and 16.5%, respectively, indicating that prolonged UV treatment leads to an increase in suboxide formation in both the bulk and interfacial regions. This observation suggests that prolonged UV treatment facilitates the formation of suboxides, implying a greater presence of uncoordinated oxygen species responsible for the generation of charge-trap states. Additionally, variations in uncoordinated oxygen species contributing to suboxide formation were examined through O 1s XPS spectra of both the bulk and interface regions of UUHL, LPSL, and HPSL, as shown in Figures 4(g)-(i) and S18, to investigate their role in the generation of charge trap states. The spectra were deconvoluted into two components: O_I at $(530.1 \pm 0.2) \text{ eV}$, corresponding to metal-oxygen (M-O) bonds, and O_{II} at $(531.1 \pm 0.2) \text{ eV}$, attributed to uncoordinated oxygen species [82-83, 86]. The relative area ratios of O_I/O_{II} for the bulk regions of UUHL, LPSL, and HPSL were 84.2/15.8%, 73.8/26.2%, and 62.8/37.2%, respectively, while those for the interface regions were 83.3/16.7%, 72.2/27.8%, and 60.2/39.2%, respectively, indicating a gradual decrease in O_I and an increase in O_{II} with longer UV treatment durations. To further verify whether the same trend persists when HfO_x is incorporated with IGZO, XPS depth profiling is presented in Figure S19. Consistent with the aforementioned observations, the relative fraction of O_{II} was found to be greater near the HfO_x /IGZO interface compared to the bulk region in both the LPSL and HPSL samples, confirming that the accumulation of uncoordinated oxygen species is slightly more highly localized near the interface region than bulk region of the HfO_x . Additionally, a slightly greater proportion of uncoordinated oxygen species was observed at the interface than in the bulk region, which can be attributed to the degradation of Hf-O bonds near the interface caused by plasma-induced damage during the sputter deposition of the channel. These results confirm that the UV treatment time increases the charge trap states in both the surface and bulk regions of HfO_x , thereby enabling distinct charge-trapping capabilities.

3.4. Mechanism of charge-trapping capability modulation in HfO_x induced by UV treatment times

The variation in the charge-trapping capability of HfO_x with different UV treatment durations can be attributed to modifications in the charge trap states. UV irradiation facilitates the dissociation of chemical bonds with lower bonding energies than the photon energy of incident UV light [87-89]. To examine whether the UV wavelengths used in this study, 185 nm (6.7 eV) and 254 nm (4.8

eV), can be absorbed by HfO_x , the optical bandgap of HfO_x was analyzed using UV–visible spectrophotometry (UV–Vis), and the corresponding Tauc plot in Figure S20 reveals a bandgap of 4.96 eV. Accordingly, 185 nm UV light possesses sufficient energy to be fully absorbed by HfO_x , while the 254 nm light can be partially absorbed due to the presence of uncoordinated species and related defect states [90].

Although the photon energies of the applied UV light are lower than the reported Hf–O bond dissociation energy of $(801 \pm 13) \text{ kJ} \cdot \text{mol}^{-1}$ ($(8.30 \pm 0.14) \text{ eV}$), the absorbed UV light is insufficient to directly dissociate Hf–O bonds within HfO_x [91]. However, it is important to note that the referenced dissociation energy corresponds to stoichiometrically stable HfO_2 . In contrast, sputter- HfO_x typically exhibits nonstoichiometric compositions, leading to the formation of unstable Hf–O bonds and being less stable with different bonding configurations and lengths, resulting in the formation of suboxide phases such as Hf_2O_3 and Hf_4O_7 , along with metal clusters and uncoordinated oxygen species, including V_O and hydroxyl groups [92–95]. To further clarify the bonding nature of sputter- HfO_x , a comparative Fourier transform infrared (FTIR) spectroscopy analysis was conducted on HfO_x films deposited by atomic layer deposition (ALD), which are widely regarded as stoichiometrically stable [96]. This analysis enables a quantitative evaluation of differences in chemical bond stiffness, with the corresponding FTIR data shown in Figure S21. In HfO_x , broad Hf–O related vibrational features are typically observed in the wavenumber range of $400\text{--}650 \text{ cm}^{-1}$ [97]. The red shift observed for the sputter- HfO_x sample indicates a lower vibrational frequency compared to ALD- HfO_x , implying reduced local Hf–O bond stiffness. The relationship between the bond stiffness and wavenumber is related to Equation 4, where ν is the vibrational wavenumber, F is the force constantly associated with the bond stiffness, and μ is the reduced mass.

$$\nu = \frac{1}{2\pi} \sqrt{\frac{F}{\mu}} \quad (4)$$

According to the relationship, this shift corresponds to a lower force constant, reflecting weakened bonding in the sputtered HfO_x film [73]. While a reduced force constant does not directly correspond to a proportional decrease in bond dissociation energy, it reflects a locally softened bonding environment associated with non-stoichiometric coordination and structural disorder. To further support this, FTIR data of HfO_x as a function of UV treatment time (10, 20, and 30 min) are presented in Figure S22, where the Hf–O vibrational frequency progressively decreases with increasing UV treatment time, confirming that prolonged UV exposure leads to a gradual reduction in bond stiffness. Such weakened Hf–O bonds exhibit increased susceptibility to further defect formation, as the reduced bond stability lowers the energy barrier for oxygen displacement under continued UV irradiation. Furthermore, UV absorption in sputter-deposited HfO_x generates additional photo-excited carriers, which drive local structural rearrangements and lattice instability, thereby providing a self-reinforcing pathway for progressive defect generation [98].

Accordingly, the mechanism of charge trap state formation in HfO_x induced by UV treatment is schematically illustrated in Figure 5(a). As determined by UV-vis spectroscopy, the optical bandgap of the sputter-deposited HfO_x is 4.96 eV, and when HfO_x is exposed to UV irradiation, photons at 184.9 nm (6.71 eV), which exceed this bandgap, are completely absorbed through band-to-band transitions, while photons at 253.7 nm (4.89 eV), which falls below the bandgap, are partially absorbed via mid-gap states, with both pathways exciting photogenerated carriers into the conduction band [99]. The excited carriers subsequently localize at intrinsic structural sites in HfO_x , forming polaron states, and upon the trapping of a second carrier at the same site, bipolaron states are created, causing significant weakening of the Hf–O bond through O–O dimer formation and structural reorganization driven by bipolaron-induced ionic displacement [100]. Thermal activation of these bipolarons, driven by the heat generated during UV irradiation, leads to Frenkel defect pair formation, generating V_O and oxygen interstitials that act as charge trap-related states within the bandgap. Since newly generated charge trap-related states tend to form preferentially near pre-existing ones, the self-amplifying formation of trap states progresses with increasing UV treatment time [101]. As the UV treatment duration increases, the density of the charge trap states in HfO_x is expected to rise due to the formation of trap states Hf–O bonds, thereby allowing a greater amount of charge to be trapped, as demonstrated in the LPSL and HPSL shown in Figure 5(b). Consequently, the charge-trapping-based

operating mechanisms of the LST and HST, which exhibit distinct charge-trapping capabilities, are schematically illustrated in Figure 5(c). When $V_{GS} > 0$ V, electrons in IGZO accumulate and become trapped within the charge trap sites of the LPSL and HPSL, leading to a reduction in channel conductance; this phenomenon is more significant in the HPSL owing to its higher trap density, which traps a larger number of electrons. Conversely, when $V_{GS} < 0$ V, the trapped electrons are released back into the IGZO channel or recombine via hole injection, increasing the channel conductance. Consequently, the HST, with its higher density of trap sites, exhibits greater conductance variation than the LST under identical bias conditions, thereby inducing distinct ΔV_{TH} values that enable differentiated conductance modulation, which is essential for implementing a region-dependent artificial pain perception system.

3.5. Emulation of nociceptive characteristics through differential conductance modulation for region-dependent pain perception

To construct a region-dependent artificial pain perception system, synaptic devices capable of differential conductance modulation were developed to emulate nociceptive characteristics that exhibit distinct pressure pain thresholds across different body regions. Hereafter, the pressure pain threshold is abbreviated as the pain threshold. These pain thresholds originate from differences in the number of activated nociceptive fibers, which give rise to variations in action potential intensity. By employing LST and HST with differential conductance modulation, region-dependent nociceptive characteristics were emulated through distinct PSC responses. The LST and HST were designed to represent body regions with low and high numbers of activated nociceptive fibers, respectively, corresponding to higher and lower pressure pain thresholds that determine sensitivity to stimuli.

PSC variations under spike intensities of -10 , -11 , -12 , -13 , and -14 V applied for 10 ms at a read voltage of 2.5 V were measured to verify the different pain thresholds of the LST and HST devices, as shown in Figure 6(a). Negative gate spikes are employed because increasing the conductance of the n-type IGZO channel requires the migration of electrons trapped in the LPSL or HPSL into the channel. The applied negative gate spike intensity drives these trapped electrons toward the channel, thereby enhancing carrier accumulation and channel conductance [102-104]. The LST exhibited PSC values of 0.28, 0.70, 1.05, 1.67, and 2.23 μ A, while the HST showed corresponding values of 0.50, 1.06, 1.76, 2.49, and 3.26 μ A, indicating that the HST exhibited higher PSC responses under identical spike intensities due to a larger number of trapped charges contributing to channel conduction compared with the LST. Furthermore, this behavior can be interpreted as a variation in the receptor potential generated in nociceptor-mediated high-threshold nociceptors under stimulation, where a higher PSC value at the same spike intensity corresponds to the generation of a higher receptor potential. In high-threshold nociceptors, pain perception is governed by a defined threshold: stimuli that generate receptor potentials exceeding this threshold are perceived as painful, whereas those that do not exceed the threshold are not recognized as pain [105-106]. Owing to its lower pain sensitivity, the LST is assigned a higher pressure pain threshold than the HST, thereby requiring a higher spike intensity for the integrated nociceptive response to reach the pain-perception state. Accordingly, the PSC responses obtained under different spike intensities demonstrate both regimes: (i) a sub-threshold regime, in which the integrated nociceptive response remains below the pain perception threshold, and no pain is perceived, and (ii) a supra-threshold regime, in which the threshold is exceeded, leading to pain perception. The differences in the pressure pain threshold between the LST and HST, which are designed to emulate the back of the hand and the calf, respectively, are attributed to differences in the number of activated nociceptors, which leads to variations in the spatial summation of nociceptive input. Accordingly, compared with the HST, the higher-pressure pain threshold of the LST implies that a greater magnitude of input stimulus is required to elicit pain perception, whereas the HST, with a greater number of activated nociceptors under identical stimulus conditions, transmits a greater pain signal than the LST does. To emulate these region-dependent differences, the threshold and magnitude of the input voltage at which firing occurred were set differently for the LST and HST, with threshold values defined as 2.0 μ A for the LST and 1.8 μ A for the HST, with threshold crossings occurring at -14 V and -13 V, respectively.

Under repetitive stimulation, LST and HST devices with different pain thresholds were used to analyze the activation of pain and the perceived pain level of nociceptors through PSC variations. Cumulative spike inputs of 3, 6, 9, and 12 within 1 s were applied using spike intensities of -13 V for the LST and -12 V for the HST, which correspond to voltages below their respective thresholds, with a pulse duration of 10 ms and a read voltage of 2.5 V, as presented in Figure S23, while Figure 6(b)–(c) visualize the corresponding PSC variations as color maps according to the number of applied pulses. As shown in the color maps, both the LST and HST exhibited a gradual increase in PSC with increasing number of input spikes, visualized by a color transition from blue, representing relatively low PSC values, to red, corresponding to higher PSC levels. This behavior, which reflects an increase in the receptor potential generated in nociceptors under continuous stimulation, becomes more evident as the number of spikes increases, implying shorter intervals between inputs and leading to cumulative activation. As the receptor potential increases, threshold crossing occurs at a spike number of 9 for the LST and 3 for the HST, resulting in pain perception once the threshold is exceeded, with the HST exhibiting higher sensitivity. The subsequent increase in PSC following threshold crossing corresponds to an increased perceived level of pain.

Following the verification of distinct pain thresholds in the LST and HST, the “no adaptation” behavior of nociceptors was further investigated, as it represents a critical protective mechanism in biological systems. Unlike visual or olfactory receptors that exhibit adaptation and consequently reduced sensitivity under continuous stimulation, nociceptors maintain their responsiveness to ensure the sustained perception of stimuli [107]. The “no adaptation” characteristic was investigated in both the LST and HST by applying 20 consecutive spike inputs of -14 V, which served as a common stimulus capable of exceeding the threshold in both devices, for 10 ms at a read voltage of 2.5 V, as shown in Figure 6(d). Once the threshold was exceeded, the pain signal was further amplified with each applied spike, reflecting the biological phenomenon in which sustained stimulation led to an intensified perception of pain. The corresponding PSC variations before and after the spike application are presented in Figure 6(e), where the PSC of the LST increased from 2.26 to 2.93 μA , and that of the HST increased from 3.28 to 4.26 μA without response attenuation. This continuous increase in the PSC confirms that sustained activation occurs without adaptation, the perceived pain intensity progressively increases and that identical inputs can evoke different pain magnitudes depending on the body region. This continuous increase in the PSC confirms sustained activation without adaptation, indicating that the perceived pain intensity continues to increase even after the threshold is exceeded, and that identical inputs can be perceived with different pain intensities depending on the body region.

Sensitization is another protective mechanism of nociceptors that emerges after tissue injury caused by stimuli exceeding the threshold, serving to minimize additional damage. Once injury occurs following a threshold-exceeding stimulus, subsequent stimuli can trigger higher pain signals even at lower intensities and a reduced pain threshold compared with the uninjured state, serving as a protective response of the body against further stimulation [108]. This abnormal pain response is closely associated with phenomena such as allodynia and hyperalgesia [109–110]. When single gate spikes of -11 V and -14 V are applied independently, the measured PSC values for the LST are 0.70 and 2.23 μA , respectively, while those for the HST are 1.06 and 3.26 μA , in agreement with the PSC responses shown in Figure 6(a). Based on these reference responses, the sensitization characteristics were examined by first applying a -14 V, which is sufficient to activate the threshold, followed by a -11 V, which does not induce threshold crossing when applied alone. The inter-spike delay was varied (0, 30, and 60 ms) to evaluate the temporal dependence of the sensitization effect. Each pulse was applied for 10 ms with a read voltage of 2.5 V, and the corresponding schematic diagram illustrating the pulse sequence along with the resulting PSC variations is presented in Figure 6(f). The LST exhibited PSC values of 1.83, 1.36, and 0.97 μA , while the HST showed corresponding values of 2.68, 2.04, and 1.73 μA for delays of 0, 30, and 60 ms, respectively. When stimulated with -11 V, which is insufficient to reach the threshold level, both the LST and HST in the injured state induced by the preceding -14 V pulse exhibited higher PSC values compared with the responses obtained under an independent -11 V stimulation. As the interval between the two spikes increased, the PSC gradually decreased, indicating a temporal reduction in activation by the prior stimulus. Although the LST did not exceed its threshold value of 2.0 μA , a noticeable increase in the PSC

was still observed, while the HST maintained a response above its threshold up to a delay of 30 ms. These results demonstrate that the devices successfully emulate the abnormal stimulus responses occurring in the injured state, thereby replicating the biological mechanism by which the body minimizes tissue damage under persistent noxious stimulation.

3.6. Implementation of a region-dependent artificial pain perception system and emulation of the sensitization phenomena of nociceptor

A region-dependent artificial pain perception system was implemented using LST and HST with distinct pressure pain thresholds, achieved through differential conductance modulation, to mimic the different numbers of activated nociceptive fibers in the back of the hand and the calf. The region-dependent artificial pain perception system is composed of a force-sensitive resistor (FSR) sensor, an Arduino-based signal processor, and synaptic transistors. As schematically illustrated in Figure 7(a), the FSR is employed to represent variations induced by external stimuli, while the Arduino-based signal processor emulates receptor interfaces that convert externally applied physical signals into biologically relevant electrical signals. These two components, which jointly mediate stimulus sensing and signal transduction, are therefore collectively defined as artificial stimulus–receptor interfaces. Furthermore, since the synaptic characteristics of the system are deliberately engineered to reflect region-dependent differences in pain perception arising from variations in the number of activated nociceptive fibers in biological systems, the LST/HST devices can be appropriately defined as artificial nociceptors. The operating principles of the region-dependent artificial pain perception system and the electrical signal conversion process of each component are illustrated in Figure S24, with the corresponding system block diagram presented in Figure S25. The FSR sensor resistance change is converted into a digital value via Arduino’s built-in ADC, which is then reconverted to an analog signal through the DAC. The op-amp subsequently amplifies and inverts the signal, with the TMA ensuring a stable current output, which is finally applied to the gate of the synaptic transistor. To characterize the resistance variation of the FSR sensor as a function of applied weight, a circuit was configured, as shown in Figure S26(a), and the resulting resistance changes are presented in Figure S26(b). The resistance decreases significantly and nearly linearly with increasing weight, which served as the basis for selecting 50, 100, and 150 g as the representative weight points. The corresponding FSR values and the magnitudes of the converted voltage pulse signals for each applied weight are summarized in Table S3, while the Arduino code detailing the conversion of FSR values into voltage pulse signals is provided in Figure S27.

Based on the system configuration, sensitization-related pain responses, including allodynia and hyperalgesia, are emulated and conceptually illustrated in Figure 7(b). These phenomena arise through a sensitization process that evolves over time, in which sustained and repeated intense stimulation induces a transition from a normal state to a sensitized state. This transition is accompanied by increased pain sensitivity and a reduced pain threshold compared with the normal condition. Allodynia is defined as the perception of pain elicited by stimuli that do not evoke pain in the normal state, whereas hyperalgesia refers to an enhanced pain response in which an identical stimulus produces stronger pain perception following sensitization. By incorporating the temporal progression of stimulation and the resulting shift in pain sensitivity, the schematic representation shows key characteristics of sensitization observed in biological nociceptive processing [111].

To emulate the characteristics of hyperalgesia and allodynia through the region-dependent artificial pain perception system, PSC responses were measured in the LST and HST under applied weights of 50, 100, and 150 g, corresponding to spike intensities of −12, −13, and −14 V, respectively, each with a duration of 200 ms. Region-dependent allodynia and hyperalgesia characteristics were reproduced by defining the uninjured and injured states as conditions without stimulation and with the application of a 150 g load corresponding to −14 V for 200 ms, respectively, thereby providing suprathreshold stimuli to both the LST and HST. The experimental data are presented in Figure S28, and Figure 7(c) shows the PSC variations corresponding to the gate voltage signals converted from the applied weights. In the uninjured state, the PSC values of the LST were 1.06, 1.69, and 2.20 μA , respectively, whereas in the HST exhibited higher responses of 1.66, 2.52, and 3.25 μA , respectively, under V_{GS} of −12, −13, and −14 V. After

sensitization, an injured state was induced by a preceding high-intensity stimulus of -14 V for 200 ms, the PSC values increased to 1.38, 2.25, and 3.17 μA in the LST and to 2.24, 3.45, and 4.26 μA in the HST under the same conditions. These results indicate that allodynia occurs at lower stimulus levels, with an applied weight of 100 g corresponding to -13 V in the LST and 50 g corresponding to -12 V in the HST, demonstrating that pain can be perceived even from stimuli that would normally remain below the threshold. Furthermore, under the same high-intensity stimulus conditions of 150 g corresponding to -14 V in the LST and 100 g and 150 g corresponding to -13 V and -14 V in the HST, larger PSC variations were observed following the injured state, which is characteristic of hyperalgesia, emulating the phenomenon of perceiving greater pain intensity from identical stimuli. Beyond the emulation of hyperalgesia, the real-time performance of the region-dependent artificial pain perception system was further evaluated to assess its potential applicability in biological electronic devices. To this end, PSC variations were measured under gradient force conditions by sequentially applying 50 g (-12 V), 100 g (-13 V), and 150 g (-14 V), five times each for a total of 15 consecutive inputs, with a pulse width of 70 ms and a read voltage of 2.5 V for 10 ms, as presented in Figure S30. As the applied weight increases (voltage increases), the PSC continuously increases, with a progressively greater variation observed at higher weights. Specifically, the PSC of the LST increased from 1.06 μA upon the initial application of 50 g (-12 V) to 3.86 μA after all 15 stimulations, while that of the HST increased from 1.38 μA to 7.43 μA . To further validate the real-time performance of the system, the average response times of the LST and HST were calculated under an applied weight of 50 g corresponding to a gate voltage spike of -12 V, as presented in Figure S29, yielding average response times of 81.9 ms and 81.8 ms, respectively, with no significant deviation observed, confirming that the system is compatible with real-time applications [112]. The distinct differences in PSC variation between the LST and HST confirm that the region-dependent artificial pain perception system is capable of real-time operation, demonstrating its potential to emulate region-dependent pain perception. Based on this observation, a comparative analysis with state-of-the-art charge trap-based artificial nociceptors is presented in Table S4, highlighting the region-dependent characteristics and distinct advantages of the proposed devices. Although the power consumption and response speed of the LST and HST are higher than those of previously reported devices, the response speed remains within the biologically relevant timescale, given that nociceptive signal transmission in the human nervous system occurs on the order of milliseconds, while the higher power consumption represents an inherent trade-off of the charge trapping-based mechanism required to achieve multi-level, region-dependent nociceptive functionality. More importantly, the primary contribution of this work lies in the demonstration of a previously unreported capability of emulating spatially distinct pain signals across different body regions by simply modulating the UV treatment duration applied to HfO_x , with successful integration with sensors and a microcontroller unit, further confirming its scalability for practical applications in robotics and neuroprosthetics [59, 113-122]. Thus, the proposed system successfully demonstrated the biomimetic emulation of nociceptors across different body regions with distinct pain thresholds, effectively emulating region-dependent pain perception characteristics. However, although UV treatment, the most critical technique employed in this study, ensures scalability through compatibility with the exposure step in conventional photolithography processes, its application beyond single devices toward fully integrated real-world products raises an important concern. Specifically, the various elements positioned beneath the device stack, such as sensors, flexible substrates, actuators, and driver circuits, may undergo unintended changes in their properties upon UV exposure, and strategies to mitigate this issue should be addressed in future work. Furthermore, for practical integration into applications such as e-skin and neuroprosthetics, arraying the sensor-transistor units will be necessary, as each transistor must independently read the resistance signal from its spatially corresponding sensor to enable region-dependent nociceptive mapping across a surface. In addition, the interface elements transmitting signals between the sensor and transistor must also exhibit stable and noise-resilient signal transfer characteristics under repeated operation [122-123].

4. Conclusions

Our findings demonstrate that varying the UV treatment time controls the trap density in both the interface and bulk regions of HfO_x , leading to differences in charge-trapping behavior that modulate the conductance characteristics and thereby enable the realization of the LST and HST. Through the modulation of conductance, the back of the hand and the calf, which possess distinct

pressure pain thresholds due to variations in the number of activated nociceptive fibers, were emulated, demonstrating no adaptation and sensitization characteristics with different firing points in terms of stimulus magnitude and frequency. This approach successfully implements the biological functionality of region-dependent pain sensitivity in the human body. Based on these findings, various synaptic functions, including spike intensity and duration dependence, SFDP, and LTP/LTD, were successfully demonstrated in the LST and HST. In particular, under identical spike intensity and duration, the LTP/LTD measurements revealed conductance changes of 134.0% and 261.0% in the device-to-device variability in the LST and HST, respectively, after 64 consecutive positive and negative spikes. These behaviors were accompanied by PSC modulation induced by charge trapping and de-trapping mechanisms, confirming distinct conductance variation characteristics under identical spikes. The PSC variations derived from the distinct characteristics of the LST and HST enabled the emulation of region-dependent nociceptor behaviors. The thresholds were set to 2.0 μ A for the LST and 1.8 μ A for the HST, and these different sensitivities allowed the demonstration of key nociceptive features such as no adaptation and sensitization, which are essential for reproducing region-dependent pain responses. Furthermore, by integrating a pressure-sensitive FSR sensor and an Arduino-based signal processor with synaptic transistors, a hardware-level region-dependent artificial pain perception system was implemented, effectively mimicking allodynia and hyperalgesia consistent with region-dependent pain transmission in the human body. Based on this functionality, realistic sensory feedback analogous to the human body was achieved, providing a simple hardware platform for customizable artificial pain perception. Achieving these requirements, including arraying of sensor-transistor units, low-voltage operation, and mechanical flexibility, would render the proposed devices fully compatible with e-skin, robotics, and neuroprosthetics applications. These findings establish the potential for e-skin, robotics, and neuroprosthetics, including therapeutic devices for pain signal modulation and artificial limbs or robots capable of perceiving region-dependent sensitivity similar to that of the human body. Furthermore, the fabrication processes involved in device manufacturing are technologies already adopted in industry, suggesting that there is sufficient potential for scalability toward large-area deposition and industrial application.

Conflict of interest

The authors declare that they have no competing interests.

Supporting Information

Supporting Information is available from the IOP Science Library or from the author.

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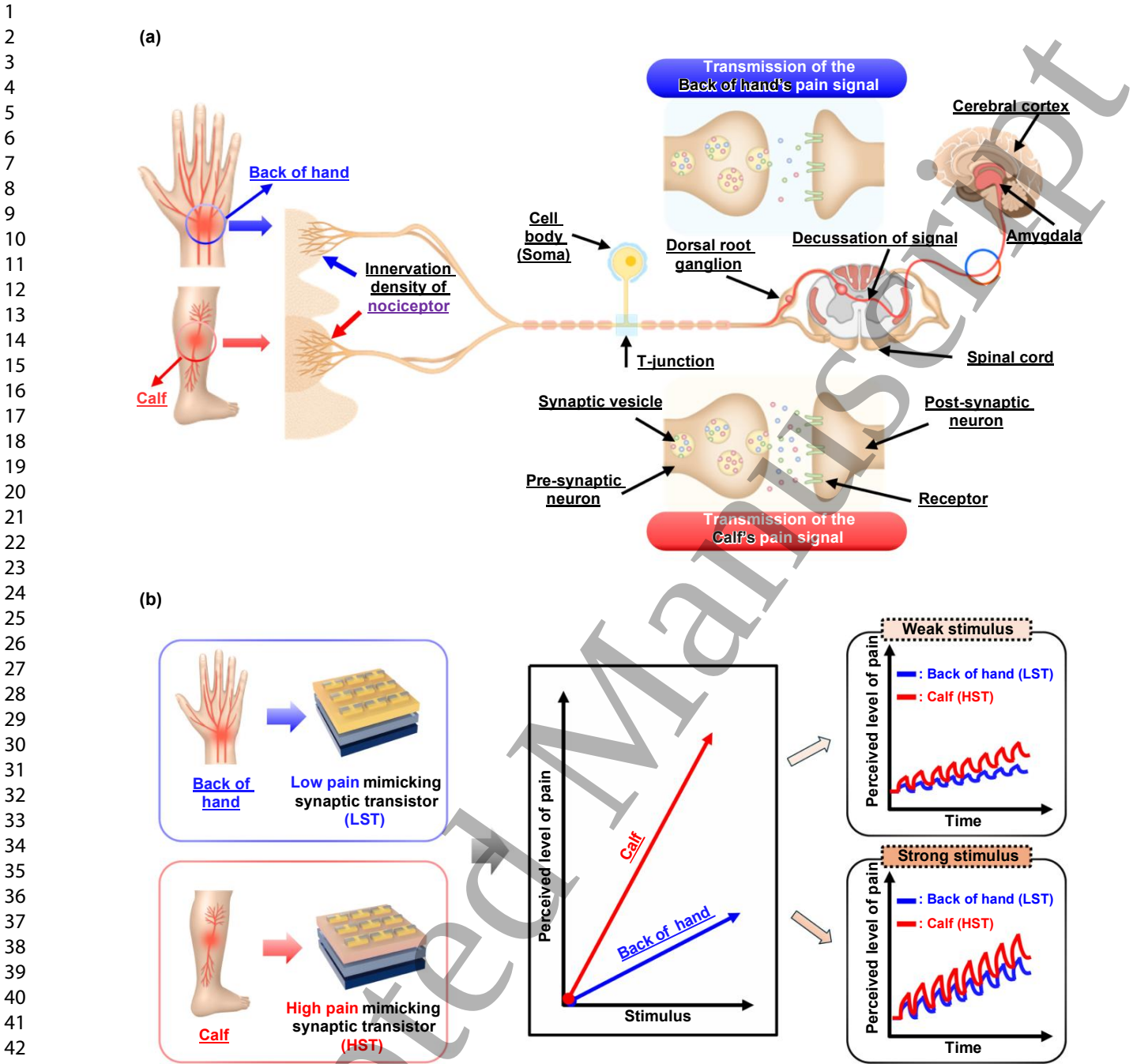


Figure 1. Schematic illustration of distinct pain sensitivities in the body and corresponding synaptic devices emulating these nociceptive responses. (a) Pain signal transmission mechanism reflecting differences in the number of activated nociceptive fibers between the back of the hand and the calf. (b) Emulation of the nociceptive characteristics of the back of the hand and the calf using the LST and HST, illustrating differences in perceived pain levels as a function of stimulus intensity.

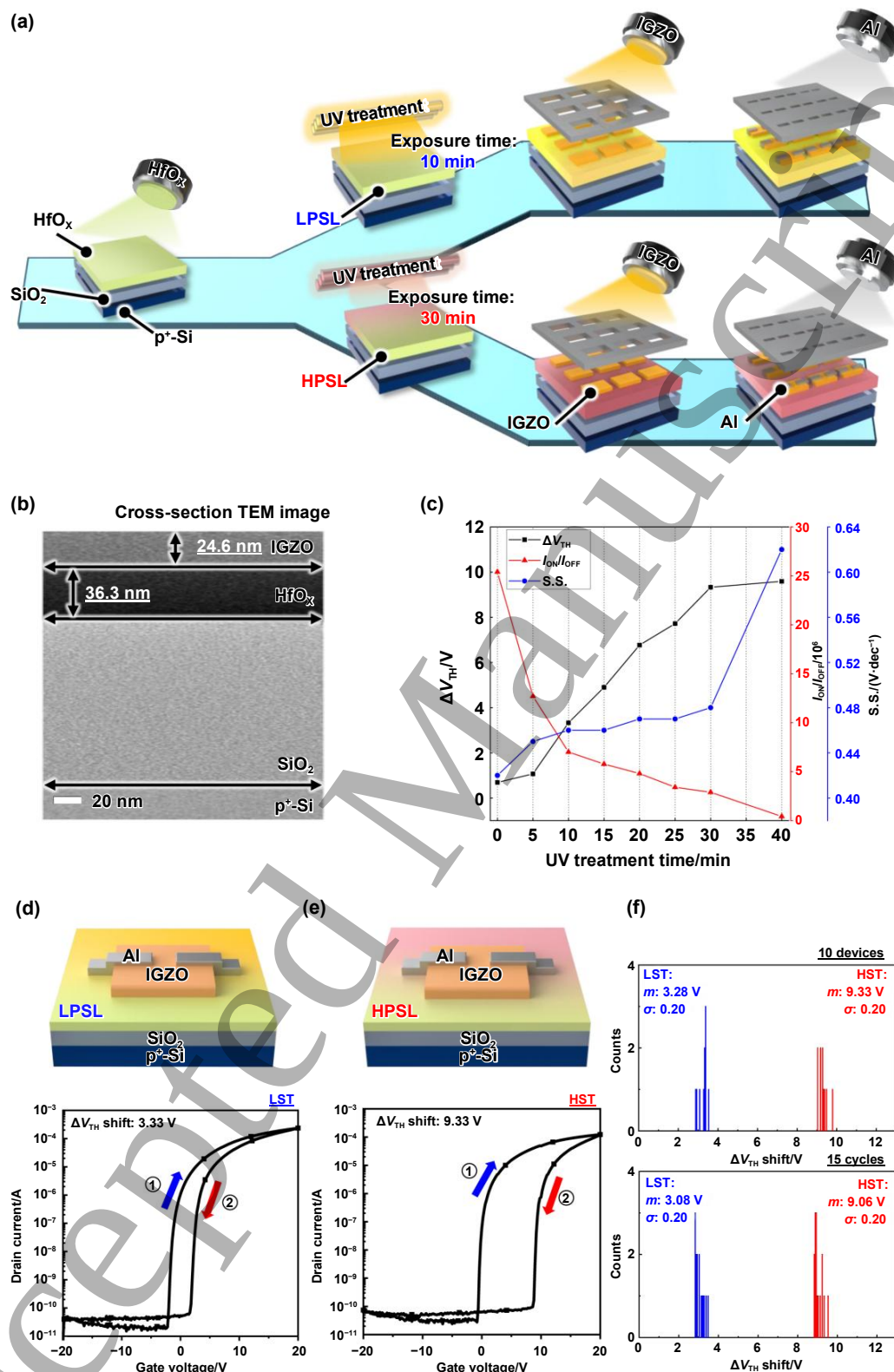


Figure 2. Fabrication procedure, structural analysis, and electrical characteristics of PSL-based synaptic transistors. (a) Schematic of the fabrication process and (b) HR-TEM cross-sectional image of PSL-based synaptic transistors. (c) Comparison of ΔV_{TH} , S.S., and I_{ON}/I_{OFF} values depending on the UV treatment time of HfO_x-based transistors. Schematic illustration of the device structure and hysteresis characteristics of the (d) LST and (e) HST. (f) Statistical results of the mean and standard deviation of ΔV_{TH} shift obtained from ten devices ($N = 10$) and fifteen consecutive cycles of the LST and HST.

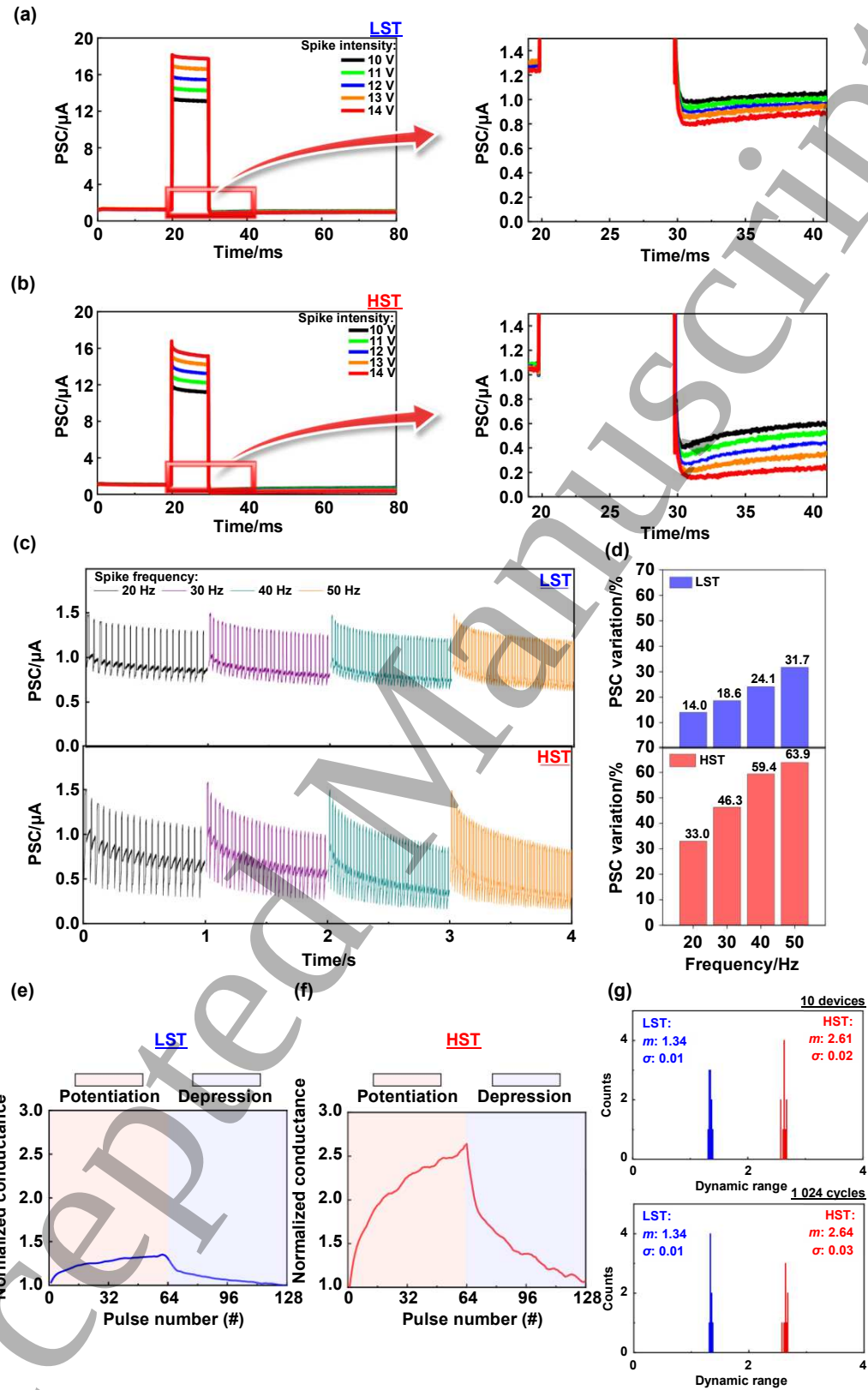


Figure 3. Synaptic characteristics of the LST and HST under a drain bias of 0.5 V. PSC responses as a function of spike intensity at a fixed duration of 10 ms for (a) the LST and (b) HST. (c) SFDP behavior and (d) corresponding PSC variations as a function of frequency ranging from 20 Hz to 50 Hz at a spike intensity of 14 V and duration of 5 ms for the LST and HST. LTP/LTD characteristics induced by 64 negative (−14 V, 30 ms) and positive (+12 V, 10 ms) pulses of (e) LST and (f) HST. (g) Statistical results of mean and standard deviation of the dynamic range obtained from ten devices (N = 10) and 1 024 cycles of the LST and HST.

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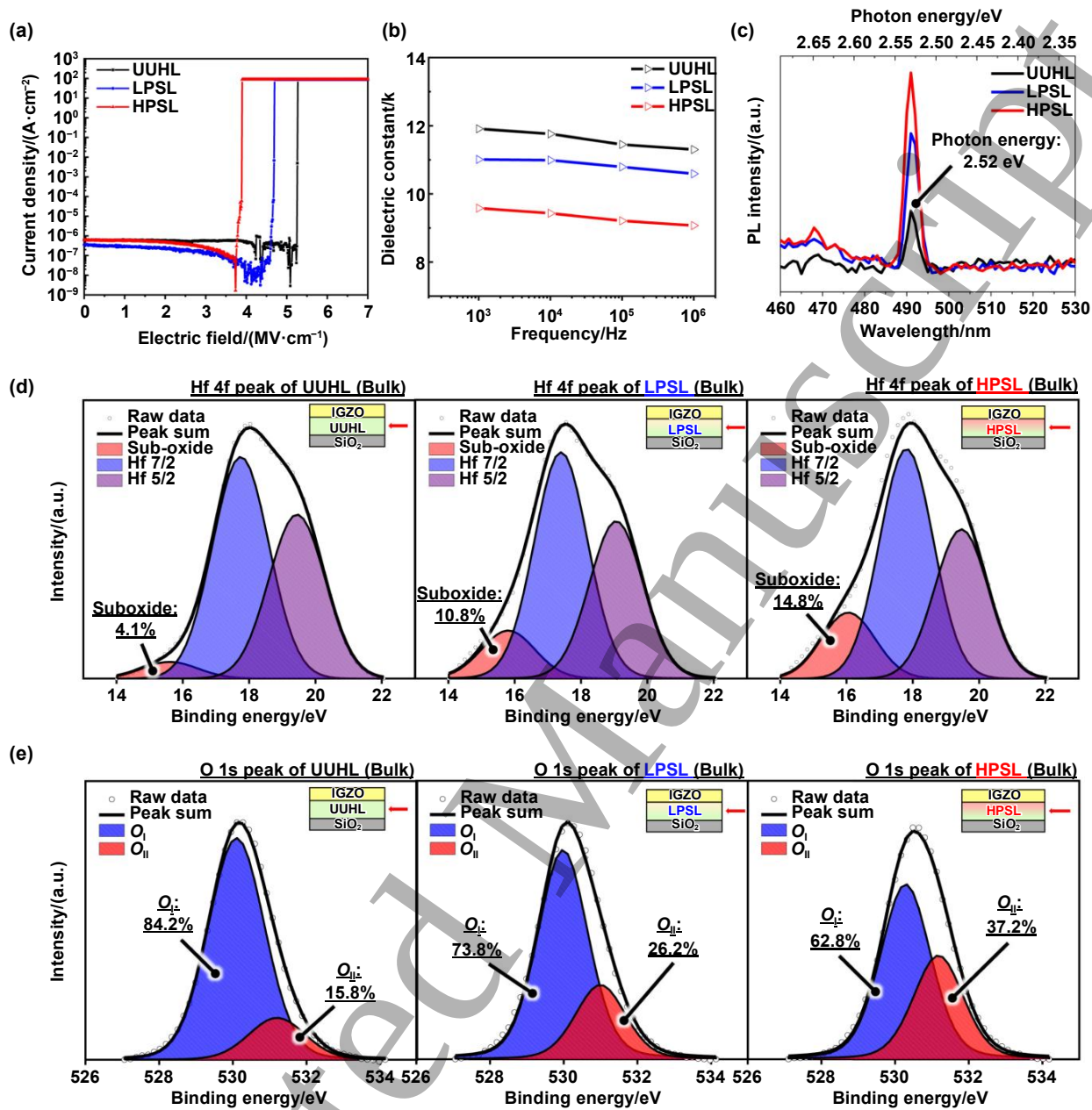


Figure 4. Analysis of trap density variation in HfO_x films as a function of UV treatment duration. Evaluation of charge trap-related state variations in UUHL, LPSL, and HPSL films through (a) J-E and (b) C-V characteristics, and identification of the energy levels of charge trap-related states by (c) PL emission spectra at 492 nm. Investigation of chemical bonding variations in the bulk regions of UUHL, LPSL, and HPSL based on deconvoluted XPS spectra: (d) Hf 4f spectra and (e) O 1s spectra.

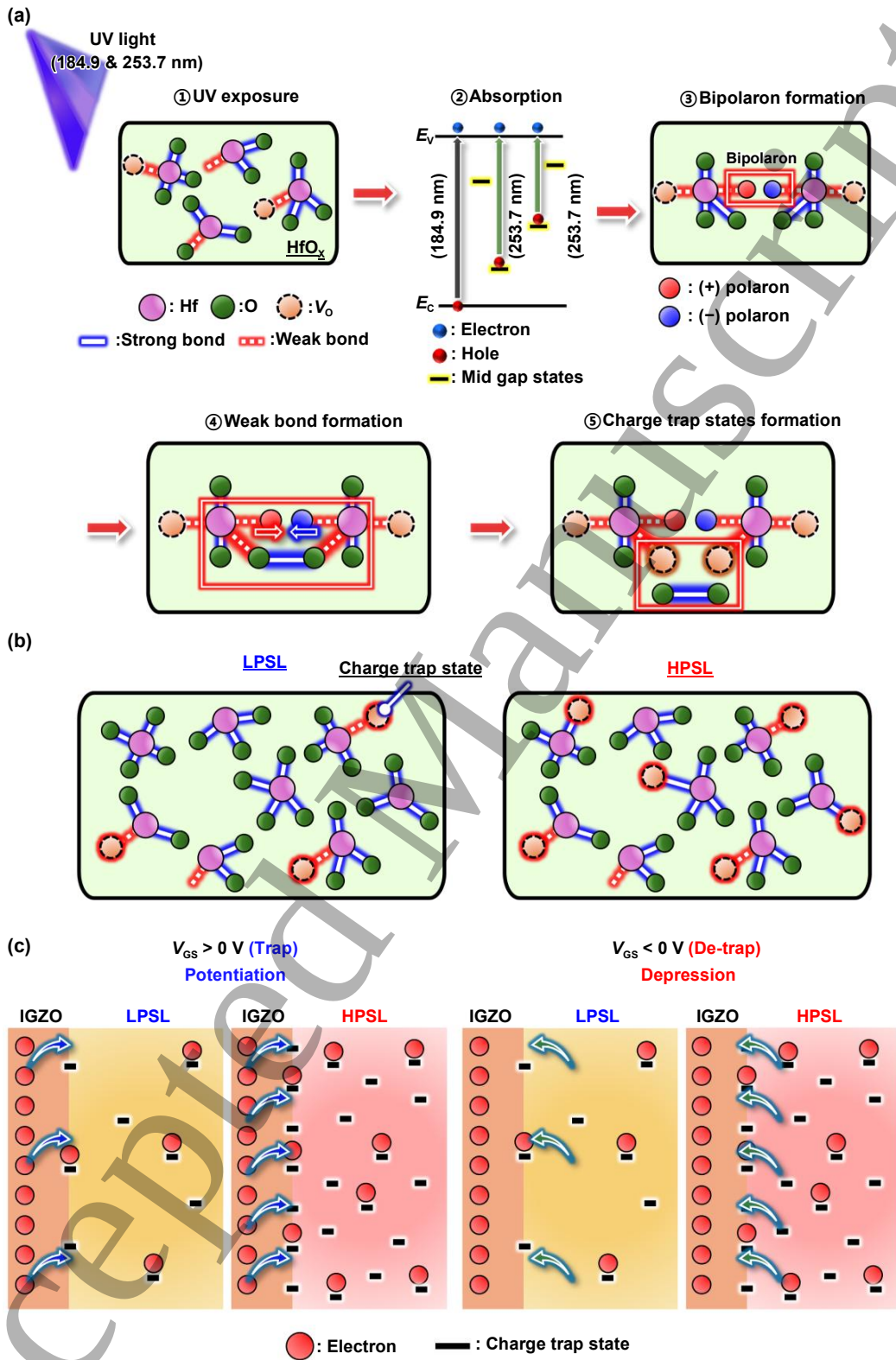


Figure 5. Schematic of charge trap density variations and operation mechanisms as a function of UV treatment time in HfO_x. (a) Schematic diagram illustrating the mechanism of charge trap state formation in HfO_x upon UV irradiation. (b) Chemical bonding variations in the LPSL and HPSL configurations with different UV treatment times. (c) Charge trap-based operation mechanisms of LPSL and HPSL between the IGZO channel, illustrating the variation in trap density under identical V_{GS} conditions.

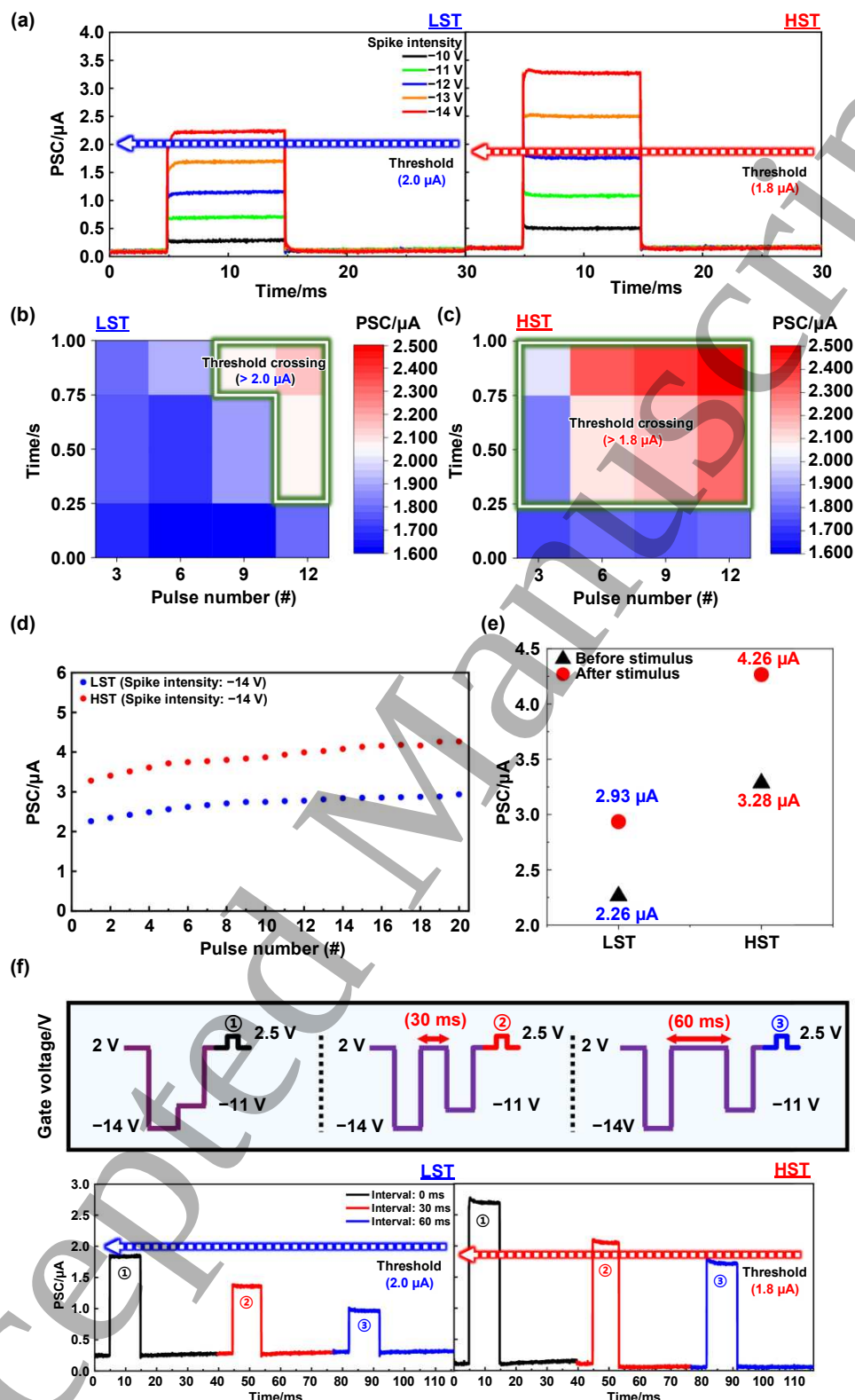


Figure 6. Nociceptive characteristics of the LST and HST for emulating region-dependent pain perception under a fixed spike intensity of 10 ms with a read voltage of 2.5 V. (a) PSC responses under varying spike intensities to confirm the different pain thresholds in the LST and HST. Color map visualizations of PSC variations with different cumulative spike numbers to identify the threshold-crossing difference for (b) LST and (c) HST. No adaptation behavior of the LST and HST was demonstrated under a spike intensity of -14 V, which simultaneously exceeds the threshold: (d) PSC responses and (e) PSC variations before and after stimulation with 20 consecutive inputs. (f) Schematic illustration of applied pulse sequences and corresponding PSC variations of the LST and HST demonstrating sensitization behavior with pulse intervals of 0, 30, and 60 ms.

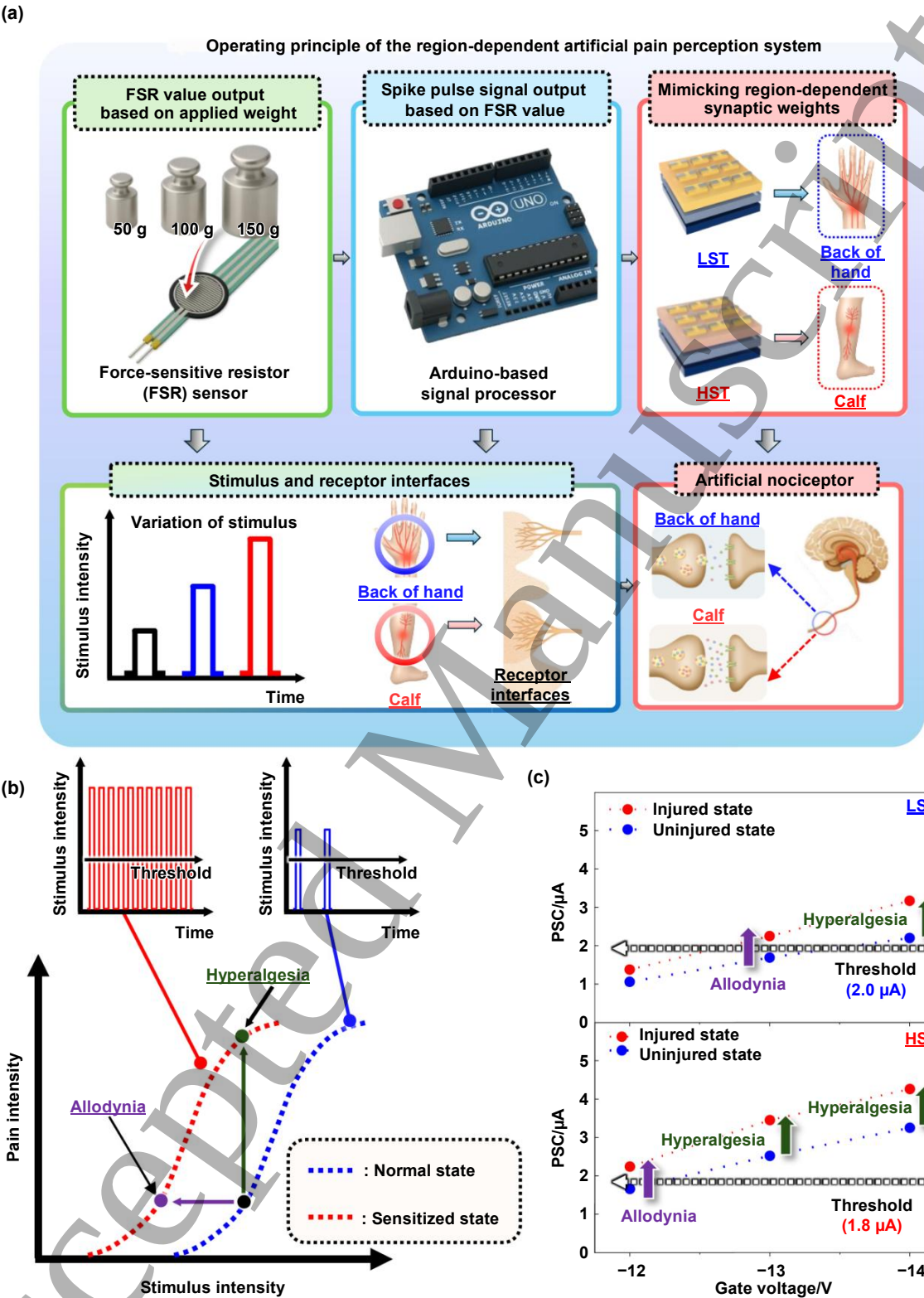


Figure 7. Emulation of a region-dependent artificial pain perception system through different pain thresholds in the LST and HST. (a) Schematic diagram and operating principle of the system composed of an FSR sensor, Arduino-based signal processor, and synaptic transistors. (b) Conceptual illustration of nociceptive sensitization incorporating temporal dynamics, showing the transition from a normal state to a sensitized state induced by sustained intense stimulation over time, resulting in allodynia and hyperalgesia through a leftward shift in the stimulus–pain intensity relationship. (c) PSC responses under applied weights of 50, 100, and 150 g, corresponding to spike intensities of -12 V, -13 V, and -14 V, demonstrate the allodynia and hyperalgesia phenomena in the LST and HST.